INDEX

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STD.: III year DIV./SEC.: AI3 SUBJECT: Computer Networks

S. No.	Date	Title	Page No.	Teacher's Sign/Remarks
1)	16/07/26	Basic Networking commands		
2)	16/07/26	Study of Different types of network cables		
3a)	23/7/26	Cisco packet Tracer environment		
3b)	23/7/26	HUB		
3b	23/7/26	SWITCH		
3b3	23/7/26	HUB & SWITCH (comparision)		
4.	30/7/26	Configure of LAN using switch		
5	30/7/26	Experiment on packet capture tool - Wireshark		
6.	6/8/26	Error detection & correction Hamming code		
7.	13/8/26	Flow control at data link layer using sliding window protocol		
8.	20/8/26	Nmap to discuss live host using Nmap scans (ARP, ICMP, TCP, UDP) on the TryHackme		
9.	27/8/26	Implementation of subnetting in cisco packet Tracer simulator		
10	27/8/26	Interneteworking with routers in CISCO Packet Tracer Simulator		
11	10/9/26	Routing at Network Layer		

S. No.	Date	Title	Page No.	Teacher's Sign/Remarks
12	17/9/26	End to End Communication a) Implement echo client using TCP b) Implement echo client using UDP c) Implement a chat program using Socket Programming.		
13	24/9/26	Ping Program		
14	8/10/26	Implement packet sniffing using raw sockets		
15	8/10/26	Analyse different types of servers using webalizer		

16/7 Exp No. 1 BASIC NETWORKING COMMANDS

Aim: Basic Networking commands in Linux and operating systems.

Basic Networking commands:

arp -a:
ARP is short form of address resolution protocol

hostname:
This is the simplest of all TCP/IP commands.

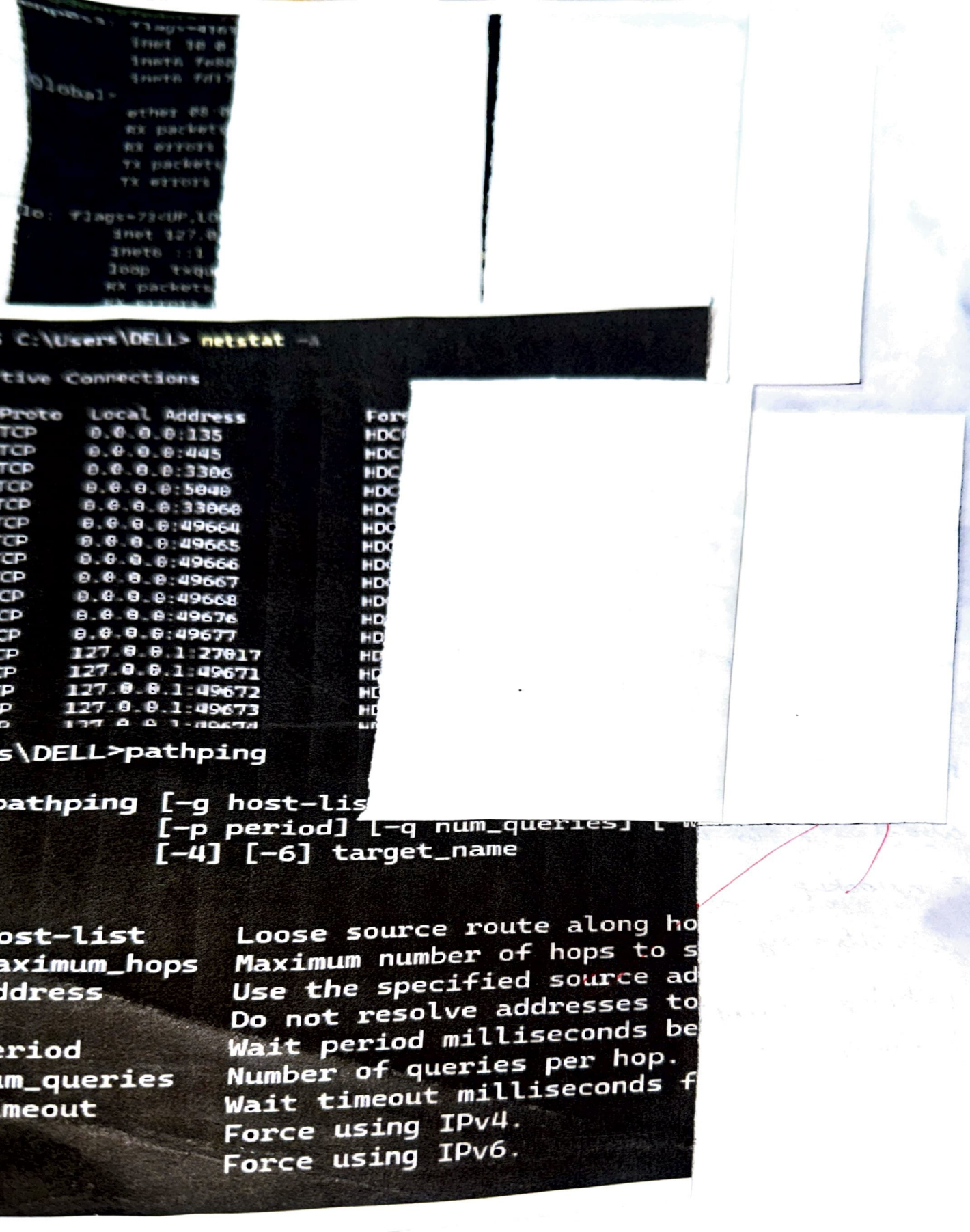
ipconfig/all:
This command displays detailed configuration information about connection including Router, Gateway, DNS, DHCP and type of Ethernet.

nbstat -a:
This command helps solve problems with NetBIOS name resolution for NetBIOS over TCP/IP

net stat:
Net stat display a variety of statistics about a Computer TCP/IP connections.

* nslookup:
It is a tool used to perform DNS lookups in display DNS details.

Pathping: Pathping is unique to windows and is basically a combination of Ttarert commands.


PS C:\Users\DELL> netstat -a
Active Connections
Proto Local Address
TCP 0.0.0.0:135
TCP 0.0.0.0:445
TCP 0.0.0.0:3306
TCP 0.0.0.0:5040
TCP 0.0.0.0:33060
TCP 0.0.0.0:49664
TCP 0.0.0.0:49665
TCP 0.0.0.0:49666
TCP 0.0.0.0:49667
TCP 0.0.0.0:49668
TCP 0.0.0.0:49676
TCP 0.0.0.0:49677
TCP 127.0.0.1:27017
TCP 127.0.0.1:49671
TCP 127.0.0.1:49672
TCP 127.0.0.1:49673
sers\DELL>pathping
: pathping [-g host-lis
[-p period] [-q num_queries]
[-4] [-6] target_name
ns:
g host-list Loose source route along ho
n maximum_hops Maximum number of hops to s
i address Use the specified source ad
n Do not resolve addresses to
p period Wait period milliseconds be
q num_queries Number of queries per hop.
w timeout Wait timeout milliseconds f
4 Force using IPv4.
6 Force using IPv6.

Route: Route command is used to show/manipulate the IP routing table.

LINUX NETWORKING COMMANDS:

ip:
The ip command is one of the basic every administrator will need in daily work. The ip command can show address information, manipulate routing, plus display network various, interfaces and tunnels.

*if config:
The if config command is a staple in many sysadmin's tool belt for configuring and troubleshooting.

*mtr:
MTR is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool.

*tcp dump:
The tcp dump command is designed for capturing and display packets.

ping: ping is a tool that verifies IP-level connectivity to another ICP/IP computer by sending internet control message Protocol (ICMP)

Result:
Basic Networking commands in both Linux and windows operating system is successfully executed and verified.